

subject to re-margining and daily mark-to-market as described in paragraph 41 (i).⁴⁴ The method is a simple approximation to Effective EPE and sets Effective EPE for a margined counterparty equal to the lesser of:

- a) Effective EPE without any held or posted margining collateral, plus any collateral that has been posted to the counterparty independent of the daily valuation and margining process or current exposure (ie initial margin or independent amount); or
- b) An add-on that reflects the potential increase in exposure over the margin period of risk plus the larger of
 - i. the current exposure net of and including all collateral currently held or posted, excluding any collateral called or in dispute; or
 - ii. the largest net exposure including all collateral held or posted under the margin agreement that would not trigger a collateral call. This amount should reflect all applicable thresholds, minimum transfer amounts, independent amounts and initial margins under the margin agreement.

The add-on is calculated as $E[\max(\Delta MtM, 0)]$, where $E[\dots]$ is the expectation (ie the average over scenarios) and ΔMtM is the possible change of the mark-to-market value of the transactions during the margin period of risk. Changes in the value of collateral need to be reflected using the supervisory haircut method or the internal estimates method, but no collateral payments are assumed during the margin period of risk. The margin period of risk is subject to the supervisory floor specified in paragraphs 41(i) to 41(iii). Backtesting should test whether realised (current) exposures are consistent with the shortcut method prediction over all margin periods within one year. If some of the trades in the netting set have a maturity of less than one year, and the netting set has higher risk factor sensitivities without these trades, this fact should be taken into account. If backtesting indicates that effective EPE is underestimated, the bank should take actions to make the method more conservative, eg by scaling up risk factor moves.

Preclude downgrade triggers from being reflected in EAD

105. In order to explicitly disallow downgrade triggers in EAD, a new paragraph 41(iv) will be inserted into Annex 4 to read as follows:

41(iv). Banks using the internal models method must not capture the effect of a reduction of EAD due to any clause in a collateral agreement that requires receipt of collateral when counterparty credit quality deteriorates.

Add requirements to improve the operational performance of the collateral department

106. To implement the requirements designed to improve the collateral department operations, two new paragraphs, 51(i) and 51(ii), will be incorporated into Annex 4 and

⁴⁴ Where a bank generally uses this shortcut method to measure Effective EPE, this shortcut method may be used by a bank that is a clearing member in a CCP for its transactions with the CCP and with clients, including those client transactions that result in back-to-back trades with a CCP.

paragraph 777(x), Part 3: The Second Pillar – Supervisory Review Process, will be revised as follows:

51(i). Banks applying the internal model method must have a collateral management unit that is responsible for calculating and making margin calls, managing margin call disputes and reporting levels of independent amounts, initial margins and variation margins accurately on a daily basis. This unit must control the integrity of the data used to make margin calls, and ensure that it is consistent and reconciled regularly with all relevant sources of data within the bank. This unit must also track the extent of reuse of collateral (both cash and non-cash) and the rights that the bank gives away to its respective counterparties for the collateral that it posts. These internal reports must indicate the categories of collateral assets that are reused, and the terms of such reuse including instrument, credit quality and maturity. The unit must also track concentration to individual collateral asset classes accepted by the banks. Senior management must allocate sufficient resources to this unit for its systems to have an appropriate level of operational performance, as measured by the timeliness and accuracy of outgoing calls and response time to incoming calls. Senior management must ensure that this unit is adequately staffed to process calls and disputes in a timely manner even under severe market crisis, and to enable the bank to limit its number of large disputes caused by trade volumes.

51(ii). The bank's collateral management unit must produce and maintain appropriate collateral management information that is reported on a regular basis to senior management. Such internal reporting should include information on the type of collateral (both cash and non-cash) received and posted, as well as the size, aging and cause for margin call disputes. This internal reporting should also reflect trends in these figures.

777(x). The bank must conduct an independent review of the CCR management system regularly through its own internal auditing process. This review must include both the activities of the business credit and trading units and of the independent CCR control unit. A review of the overall CCR management process must take place at regular intervals (ideally not less than once a year) and must specifically address, at a minimum:

- the adequacy of the documentation of the CCR management system and process;
- the organisation of the collateral management unit;
- the organisation of the CCR control unit;
- the integration of CCR measures into daily risk management;
- the approval process for risk pricing models and valuation systems used by front and back-office personnel;
- the validation of any significant change in the CCR measurement process;
- the scope of counterparty credit risks captured by the risk measurement model;
- the integrity of the management information system;
- the accuracy and completeness of CCR data;
- the accurate reflection of legal terms in collateral and netting agreements into exposure measurements;

- the verification of the consistency, timeliness and reliability of data sources used to run internal models, including the independence of such data sources;
- the accuracy and appropriateness of volatility and correlation assumptions;
- the accuracy of valuation and risk transformation calculations; and
- the verification of the model's accuracy through frequent backtesting.

Requirements on the controls around the reuse of collateral by IMM banks

107. To implement the requirements on controls regarding the reuse of collateral, a new paragraph 51(iii) will be included in Annex 4 as follows:

51(iii). A bank employing the internal models method must ensure that its cash management policies account simultaneously for the liquidity risks of potential incoming margin calls in the context of exchanges of variation margin or other margin types, such as initial or independent margin, under adverse market shocks, potential incoming calls for the return of excess collateral posted by counterparties, and calls resulting from a potential downgrade of its own public rating. The bank must ensure that the nature and horizon of collateral reuse is consistent with its liquidity needs and does not jeopardise its ability to post or return collateral in a timely manner.

Require banks to use supervisory haircuts when transforming non-cash OTC collateral into cash-equivalent.

108. To implement the supervisory haircuts for non-cash OTC collateral, a new paragraph 61(i) would be incorporated in Annex 4 as follows:

61(i). For a bank to recognise in its EAD calculations for OTC derivatives the effect of collateral other than cash of the same currency as the exposure itself, if it is not able to model collateral jointly with the exposure then it must use either haircuts that meet the standards of the financial collateral comprehensive method with own haircut estimates or the standard supervisory haircuts.

Requirement for banks to model non-cash collateral jointly with underlying securities for OTC Derivatives and SFTs.

109. To ensure the robustness of non-cash collateral, a new paragraph 61(ii) will be inserted in Annex 4 as follows:

61(ii). If the internal model includes the effect of collateral on changes in the market value of the netting set, the bank must model collateral other than cash of the same currency as the exposure itself jointly with the exposure in its EAD calculations for securities-financing transactions.

Revise credit risk mitigation section to add a qualitative collateral management requirement

110. To ensure that sufficient resources are devoted to the orderly operation of margin agreements for OTC derivative and SFT counterparties, and that appropriate collateral management policies are in place, a new paragraph 115(i) will be inserted into the main text and will read as follows:

115(i). Banks must ensure that sufficient resources are devoted to the orderly operation of margin agreements with OTC derivative and securities-financing